

The background of the entire image is a dark blue field filled with a pattern of red dots. These dots are arranged in a way that they form a large, faint circular shape in the center, with the density of the dots being higher in the center and fading out towards the edges. The dots are of varying sizes, creating a textured, digital effect.

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Chaos Control And Schedule Of Shuttle Buses

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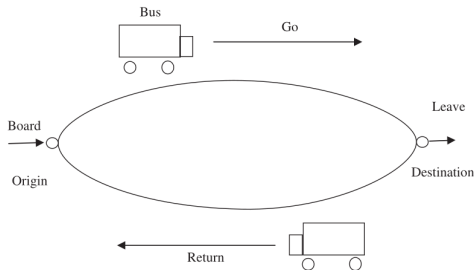
Motivation: The Bus Bunching Problem

A familiar phenomenon:

- ▶ Shuttle buses run back and forth between two terminals with no fixed timetable.
- ▶ Trip duration depends on **passengers picked up**, which depends on the **headway** since the previous bus.

The feedback loop → chaos:

1. Bus falls behind $\Rightarrow$ more passengers $\Rightarrow$ longer boarding $\Rightarrow$ falls *further* behind.
2. Next bus finds fewer passengers, speeds up.
3. Gaps amplify every round $\Rightarrow$ **deterministic chaos**.



The Model: Setup and Notation

System: M buses shuttle between Origin and Destination. Passengers board at Origin, alight at Destination.

Key assumptions:

- ▶ Passengers arrive at rate μ (constant).
- ▶ Each boarding takes γ , each alighting takes η seconds.
- ▶ Buses can *overtake* each other.
- ▶ A delayed bus can *speed up* to compensate.

Notation:

Symbol	Meaning
M	number of buses
$t_i(m)$	arrival time of bus i , trip m
$B_i(m)$	passengers boarding bus i
$V_i(m)$	average speed of bus i
μ	passenger arrival rate
γ, η	board / alight time per pax
L	one-way route length
V_0	base speed (no speedup)
s_i	speedup coefficient of bus i

The Model: Arrival Time Equation (Eq. 1)

A bus trip has three phases: **arrive** at Origin, **board/alight** passengers, **drive** the round trip. So the *next* arrival time is:

$$t_i(m+1) = \underbrace{t_i(m)}_{\text{current time}} + \underbrace{(\gamma + \eta) B_i(m)}_{\text{dwell time}} + \underbrace{\frac{2L}{V_i(m)}}_{\text{travel time}} \quad (\text{Eq. 1})$$

- ▶ $(\gamma + \eta) B_i(m)$: total time spent boarding and alighting passengers.
- ▶ $\frac{2L}{V_i(m)}$: round-trip driving time at average speed $V_i(m)$.

⇒ We need expressions for $B_i(m)$ and $V_i(m)$ to close this equation.

The Model: Passengers and Speedup (Eq. 2–3)

Eq. 2 – Passengers boarding: proportional to the headway (time since last bus).

$$B_i(m) = \mu (t_i(m) - t_{i'}(m')) \equiv \mu h_i(m) \quad (\text{Eq. 2})$$

where i' is whichever bus arrived just before bus i , and $h_i(m)$ is the **headway**.

Eq. 3 – Speedup control: a bus with more passengers drives faster to compensate.

$$V_i(m) = V_0 + s_i (\gamma + \eta) B_i(m) \quad (\text{Eq. 3})$$

- ▶ $s_i = 0$: no compensation – speed is always V_0 .
- ▶ s_i large: heavy speedup – the bus tries hard to catch up.

Two competing effects: large headway $\rightarrow$ more passengers $\rightarrow$ *longer dwell* but also *faster driving*.

The Model: Combined Raw Equation (Eq. 4)

Substituting Eq. 2 and Eq. 3 into Eq. 1:

$$t_i(m+1) = t_i(m) + \mu(\gamma + \eta)(t_i(m) - t_{i'}(m')) + \frac{2L}{V_0 + s_i \mu(\gamma + \eta)(t_i(m) - t_{i'}(m'))} \quad (\text{Eq. 4})$$

Reading this equation:

- ▶ **Dwell term** $\mu(\gamma + \eta)h$: grows linearly with headway $\rightarrow$ delays the bus.
- ▶ **Travel term** $\frac{2L}{V_0 + s_i \mu(\gamma + \eta)h}$: decreases with headway $\rightarrow$ speeds up the bus.

$\Rightarrow$ This is a **nonlinear map**: the next arrival depends on the current headway through both a linear and a reciprocal term. This competition is the source of chaos.

The Model: Dimensionless Map (Eq. 5)

Non-dimensionalise by the free-running round-trip time $\tau_0 = 2L/V_0$:

$$T_i \equiv \frac{t_i}{\tau_0}, \quad \Gamma \equiv \mu(\gamma + \eta), \quad S_i \equiv \frac{s_i \mu(\gamma + \eta) \cdot 2L}{V_0^2}$$

Defining the **headway** $H_i(m) \equiv T_i(m) - T_{i'}(m')$, we get:

$$T_i(m+1) = T_i(m) + \Gamma \cdot H_i(m) + \frac{1}{1 + S_i \cdot H_i(m)} \quad (\text{Eq. 5})$$

- ▶ Γ : **loading parameter** – how strongly passengers slow the bus.
- ▶ S_i : **speedup parameter** – how aggressively bus i compensates.
- ▶ H_i : **headway** – the time gap to the previous bus.

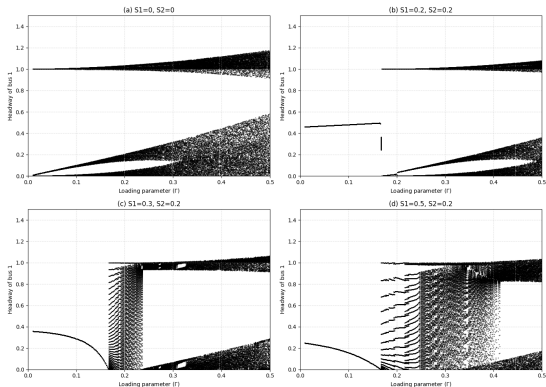
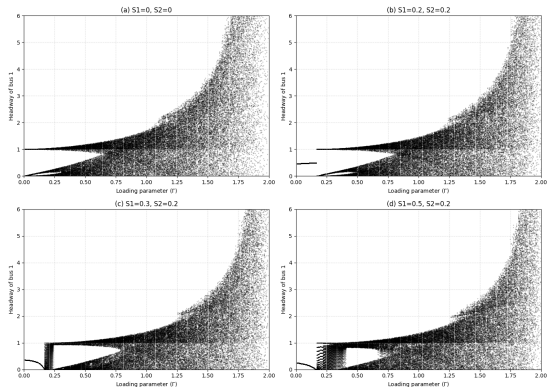
The entire dynamics depend only on Γ and $\{S_i\}$ – all physical constants absorbed.

Key subtlety: buses can *overtake* each other, so the “preceding bus” in $H_i(m)$ is **not fixed in advance** – it depends on the dynamics itself.

Simulation:

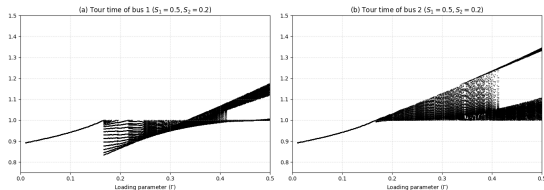
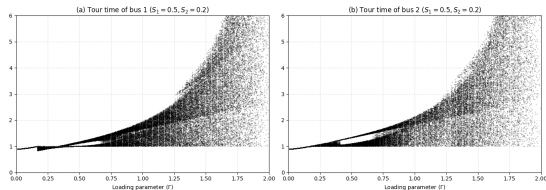
1. Maintain each bus's next arrival time on a priority queue.
2. Pop the earliest arrival – that bus has just returned to Origin.
3. Compute its headway H_i against whichever bus arrived most recently (not a pre-assigned partner).
4. Apply the nonlinear map (Eq. 5) to compute the next arrival time; push it back onto the queue.
5. Repeat for all trips; discard the first 900 (transient) and analyse trips 900–1000.

Time Series: Headway Dynamics



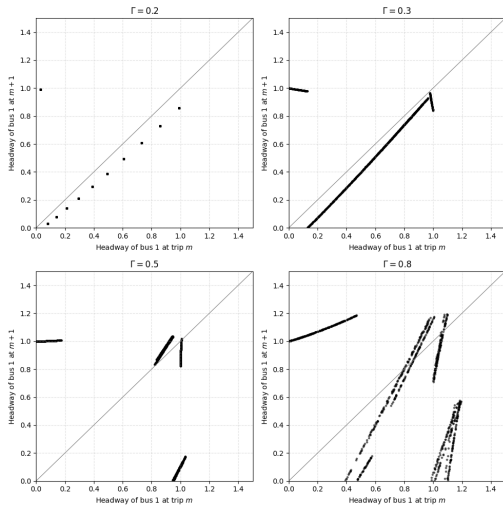
For small Γ , the headways quickly converge to a fixed value (**regular motion**). As Γ increases past Γ^* , the headway begins to oscillate **periodically**, then **chaotically**.

Time Series: Tour-Time Dynamics



Tour time mirrors headway through the map: once headway oscillates chaotically, the round-trip duration fluctuates irregularly – directly undermining **schedule reliability**.

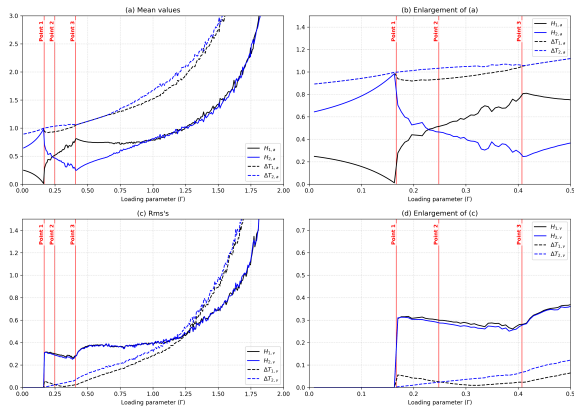
Phase Space: Return Maps



Reading the map

- ▶ $\Gamma = 0.2$: periodic motion with period 11.
- ▶ $\Gamma = 0.3$: piecewise map, chaotic motion with a single extended distribution.
- ▶ $\Gamma = 0.5$: piecewise map, chaotic motion with two extended distributions.
- ▶ $\Gamma = 0.8$: very complex piecewise map, multi-valued function, chaotic motion with a single extended distribution.

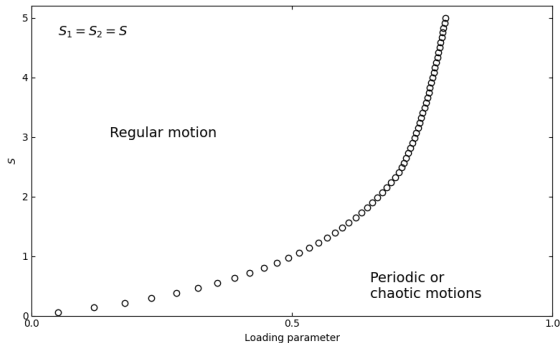
Phase Space: Mean and RMS Statistics



Reading the curves

- ▶ **RMS ≈ 0 :** for small Γ – headway converges to a constant (regular).
- ▶ **RMS jumps sharply:** at $\Gamma = \Gamma^*$ – the onset of periodic/chaotic oscillation.
- ▶ **Mean tour time** grows quickly as $\Gamma \rightarrow 2$, foreshadowing collapse.

Phase Space: The Phase Diagram

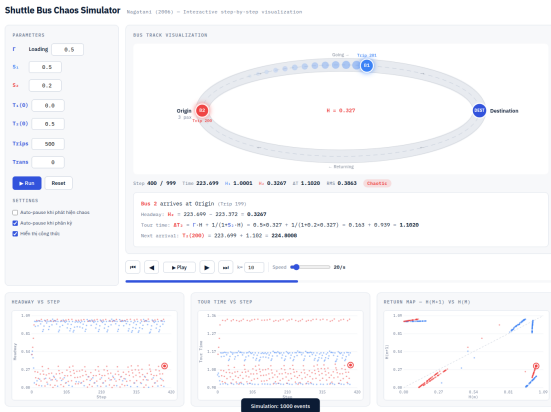


Two regimes

For the symmetric case $S_1 = S_2 = S$, there exists a critical loading $\Gamma^*(S)$ separating **regular motion** (below) from **periodic or chaotic motion** (above).

- ▶ No closed-form expression for $\Gamma^*(S)$ exists.
- ▶ The boundary is **determined numerically** by bisection search (diamonds in figure).

Interactive Demo: Live Chaos Simulator



What you can do:

- ▶ Set Γ , S_1 , S_2 and watch chaos unfold step-by-step.
- ▶ Buses animate around the track in real time, pausing to board at the Origin and alight at the Destination.
- ▶ Return-map plots update live.



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THANK YOU !

